



covalent ionic and metallic bonding

Name:

Class:

Date:

Time:

61 minutes

Marks:

60 marks

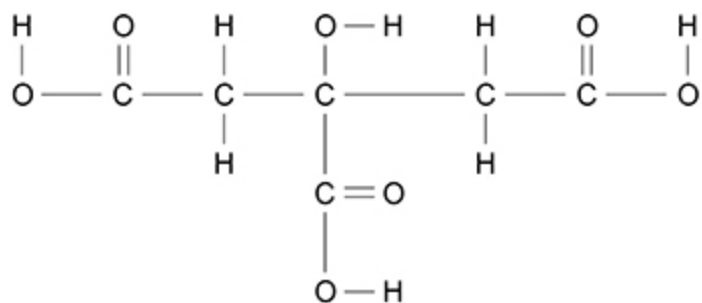
Comments:

1.

This question is about citric acid.

Figure 1 represents one molecule of citric acid.

Figure 1



- (a) Complete the molecular formula of citric acid.

Use **Figure 1**.



(1)

- (b) What type of bonding is shown in **Figure 1**?

Tick (✓) **one** box.

Covalent

☐

Ionic

☐

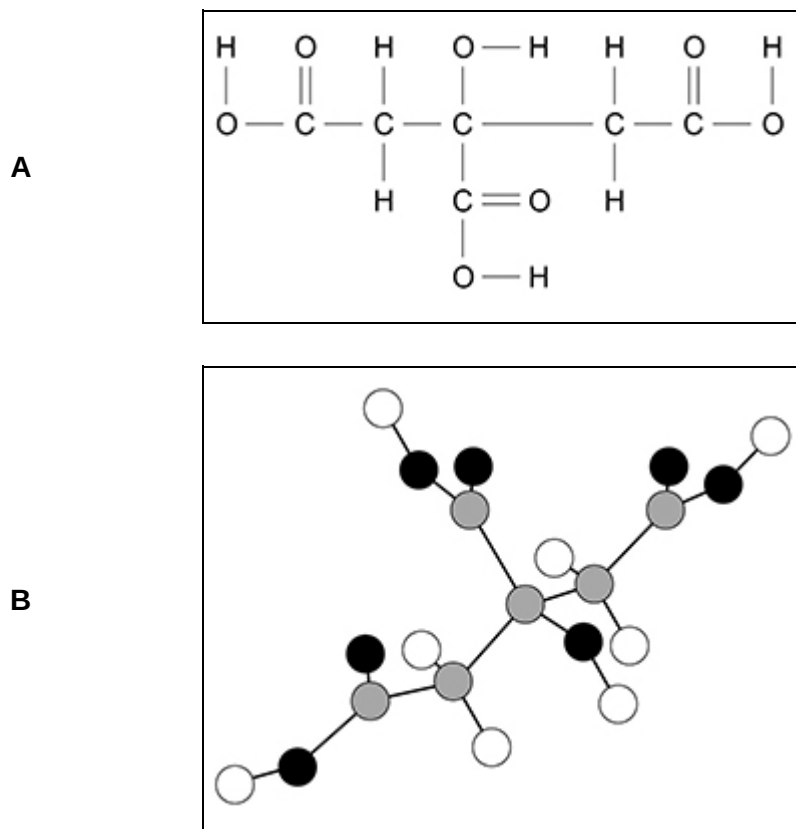
Metallic

☐

(1)

(c) **Figure 2** shows two representations of one molecule of citric acid, **A** and **B**.

Figure 2



Give **two** advantages of representation **A** compared with representation **B**.

Advantages of **A**:

1 _____

2 _____

(2)

A student investigated the temperature change during the reaction between citric acid and sodium hydrogencarbonate solution.

Citric acid is a solid.

This is the method used.

1. Pour 25 cm³ of sodium hydrogencarbonate solution into a polystyrene cup.
2. Measure the temperature of the sodium hydrogencarbonate solution.
3. Add 0.25 g of citric acid to the cup.
4. Stir the solution.
5. Measure the temperature of the solution.
6. Repeat steps 3 to 5 until a total of 2.00 g of citric acid has been added.

The table below shows some of the student's results.

Mass of citric acid added in g	Temperature of solution in °C
0.00	22.6
0.25	22.2
0.50	21.8
0.75	21.4
1.00	21.0
1.25	20.6

(d) How do the results in table above show that the reaction is endothermic?

(1)

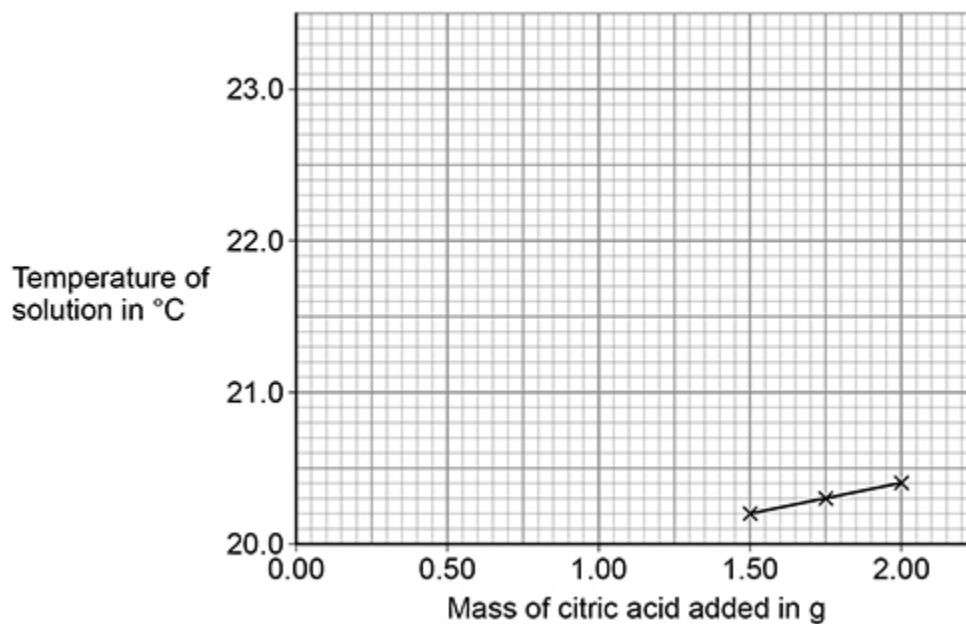
(e) Three of the student's results are plotted on the graph below.

A line of best fit for these points is drawn.

Complete the graph below.

You should:

- plot the data from table above on the graph below
- draw a line of best fit through the points you have plotted
- extend your line of best fit to meet the line of best fit already drawn on the graph below.



(4)

(f) Determine the overall temperature change for the reaction.

Use the graph above.

Overall temperature change = _____ °C

(2)

(g) What is the dependent variable in this investigation?

Tick (✓) **one** box.

Mass of citric acid

☐

Temperature of solution

☐

Volume of solution

☐

(1)

(Total 12 marks)

2.

This question is about magnesium and magnesium chloride.

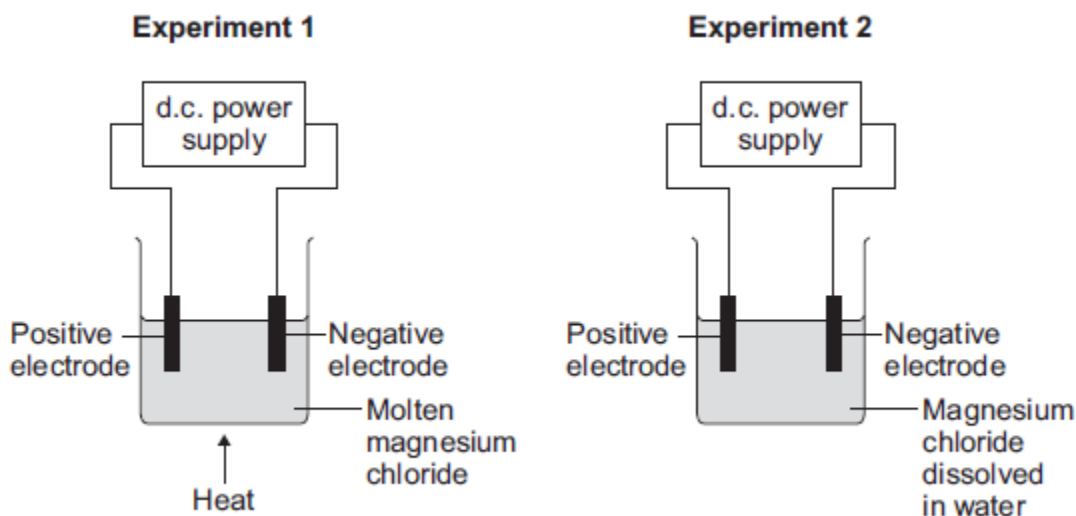
(a) Magnesium chloride contains magnesium ions (Mg^{2+}) and chloride ions (Cl^-).

Describe, in terms of electrons, what happens when a magnesium atom reacts with chlorine atoms to produce magnesium chloride.

(4)

- (b) Magnesium chloride can be electrolysed.

The diagram below shows two experiments for electrolysing magnesium chloride.



- (i) Explain why magnesium chloride must be molten or dissolved in water to be electrolysed.

(2)

- (ii) Explain how magnesium is produced at the negative electrode in **Experiment 1**.

(3)

- (iii) In **Experiment 2** a gas is produced at the negative electrode.
Name the gas produced at the negative electrode.

(1)

- (iv) Suggest why magnesium is **not** produced at the negative electrode in **Experiment 2**.

(1)

- (v) Complete and balance the half equation for the reaction at the positive electrode.



(1)

- (c) Magnesium is a metal.

Explain why metals can be bent and shaped.

(2)

(Total 14 marks)

3.

This question is about copper.

- (a) Copper can be extracted by smelting copper-rich ores in a furnace.

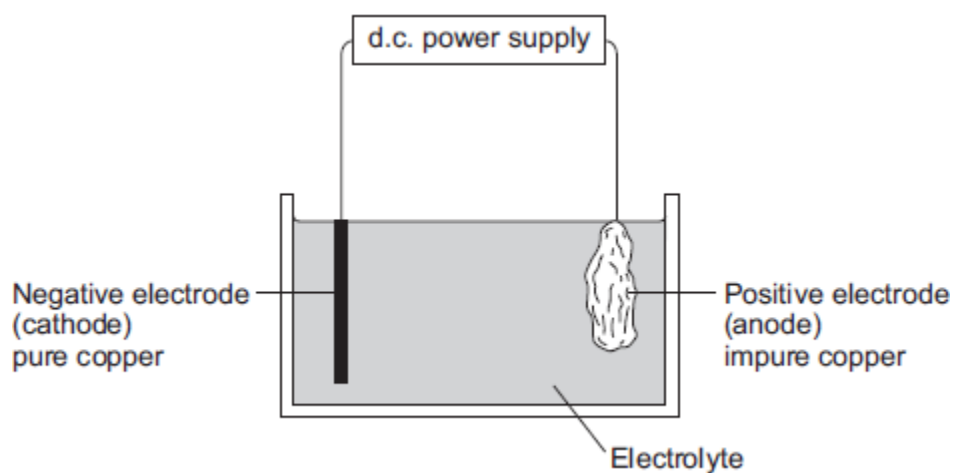
The equation for one of the reactions in the smelting process is:



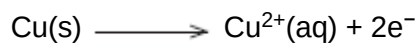
Explain why there would be an environmental problem if sulfur dioxide gas escaped into the atmosphere.

(2)

- (b) The impure copper produced by smelting is purified by electrolysis, as shown below.



Copper atoms are oxidised at the positive electrode to Cu^{2+} ions, as shown in the half equation.



- (i) How does the half equation show that copper atoms are oxidised?

(1)

- (ii) The Cu^{2+} ions are attracted to the negative electrode, where they are reduced to produce copper atoms.

Write a balanced half equation for the reaction at the negative electrode.

(1)

- (iii) Suggest a suitable electrolyte for the electrolysis.

(1)

- (c) Copper metal is used in electrical appliances.

Describe the bonding in a metal, and explain why metals conduct electricity.

(4)

- (d) Soil near copper mines is often contaminated with low percentages of copper compounds.

Phytomining is a new way to extract copper compounds from soil.

Describe how copper compounds are extracted by phytomining.

(3)

- (e) A compound in a copper ore has the following percentage composition by mass:

55.6% copper, 16.4% iron, 28.0% sulfur.

Calculate the empirical formula of the compound.

Relative atomic masses (A_r): S = 32; Fe = 56; Cu = 63.5

You must show all of your working.

Empirical formula = _____

(4)

(Total 16 marks)

4.

This question is about sodium chloride and iodine.

- (a) Describe the structure and bonding in sodium chloride.

(4)

- (b) When sodium chloride solution is electrolysed, one product is chlorine.

Name the **two** other products from the electrolysis of sodium chloride solution.

(2)

- (c) Many people do not have enough iodine in their diet.

Sodium chloride is added to many types of food. Some scientists recommend that sodium chloride should have a compound of iodine added.

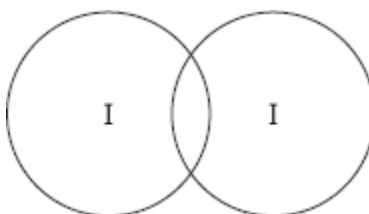
Give **one** ethical reason why a compound of iodine should **not** be added to sodium chloride used in food.

(1)

- (d) The bonding in iodine is similar to the bonding in chlorine.

- (i) Complete the diagram below to show the bonding in iodine.

Show the outer electrons only.



(2)

- (ii) Explain why iodine has a low melting point.

(3)

(iii) Explain, in terms of particles, why liquid iodine does not conduct electricity.

(2)

(Total 14 marks)

5.

Millions of years ago the Earth formed as a giant ball of molten rock. The outer surface cooled forming a thin, solid outer crust. Volcanic activity on the surface produced an atmosphere containing the compounds carbon dioxide, ammonia, methane and water vapour.

Describe the bonding in any **one** of these compounds. You must include electronic structures in your explanation.

(Total 4 marks)

Mark schemes

- 1.**
- (a) $\text{C}_6\text{H}_8\text{O}_7$ 1
- (b) covalent 1
- (c) shows (single and) double bonds 1
- shows which atoms are which element 1
- (d) temperature decreases (during the reaction)
allow (the solution) gets colder 1
- (e) all six points plotted correctly
allow a tolerance of $\pm \frac{1}{2}$ small square
allow 1 mark for four / five points plotted correctly 2
- line of best fit 1
- extrapolation to meet the printed line 1
- (f) $22.6 - 20.2$
allow ecf from question (e) 1
- $= 2.4 (^{\circ}\text{C})$
ignore sign
if no other mark awarded allow 1 mark for $2.2 (^{\circ}\text{C})$ 1
- (g) temperature of solution 1
- [12]
- 2.**
- (a) magnesium loses two electrons **and** chlorine gains one electron
*accept magnesium loses electrons **and** chlorine gains electrons for 1 mark*
ignore oxidation and reduction 2
- one magnesium and two chlorines
accept MgCl_2 1

noble gas structure

or

eight electrons in the outer shell

accept full outer shell (of electrons)

or

(electrostatic) attraction between ions

or

forms ionic bonds

*do **not** accept covalent bonds*

1

*reference to incorrect particles **or** incorrect bonding **or** incorrect structure = **max 3***

(b) (i) because ions can move

ignore ions attracted

*do **not** accept molecules / atoms moving*

*do **not** accept incorrect reference to electrons moving*

1

(and ions move) to the electrodes

or

(and ions) carry charge

1

accept converse for solid

(ii) magnesium (ions) attracted (to the electrode)

1

so magnesium ions gain electrons

accept magnesium ions are reduced

ignore oxidised

1

2 electrons

*accept a correct half equation for 2nd **and** 3rd marking points*

1

(iii) hydrogen

allow H₂

1

(iv) magnesium is more reactive than hydrogen
accept converse
*allow magnesium is high in the reactivity series **or** magnesium is very/too reactive.*
*do **not** accept magnesium ions are more reactive than hydrogen ions*

1

(v) $2\text{Cl}^- \rightarrow \text{Cl}_2 + 2\text{e}^-$
must be completely correct

1

(c) layers (of particles/atoms/ions)
 (particles/atoms/ions/layers) can slide

1

1

*any mention of intermolecular / weak bonds/forces = **max 1***

[14]

3.

(a) because sulfur dioxide causes acid rain

1

which kills fish / aquatic life **or** dissolves / damages statues / stonework **or** kills / stunts growth of trees

if no other mark awarded then award 1 mark for sulfur dioxide is toxic or causes breathing difficulties.

1

(b) (i) electrons are lost

1

(ii) $\text{Cu}^{2+} + 2\text{e}^- \rightarrow \text{Cu}$
allow $\text{Cu}^{2+} \rightarrow \text{Cu} - 2\text{e}^-$
ignore state symbols

1

(iii) copper sulfate
allow any ionic copper compound

1

(c) (lattice of) positive ions

1

delocalised electrons

accept sea of electrons

1

(electrostatic) attraction between the positive ions and the electrons

1

electrons can move through the metal / structure **or** can flow

allow electrons can carry charge through the metal / structure

if wrong bonding named or described or attraction between

oppositely charged ions then do not award M1 or M3 – MAX 2

1

(d) (copper compounds are absorbed / taken up by) plants

allow crops

1

which are burned

1

the ash contains the copper compounds

do not award M3 if the ash contains copper (metal)

1

(e)

/ A _r	55.6 / 63.5	16.4 / 56	28.0 / 32
moles	0.876	0.293	0.875
ratio	3	1	3
formula	Cu ₃ FeS ₃		

award 4 marks for Cu₃FeS₃ with some correct working

*award 3 marks for Cu₃FeS₃ with **no** working*

if the answer is not Cu₃FeS₃ award up to 3 marks for correct steps from the table apply ecf

if the student has inverted the fractions award 3 marks for an answer of CuFe₃S

4

[16]

4.

(a) lattice / giant structure

max 3 if incorrect structure or bonding or particles

1

ionic **or** (contains) ions

1

Na⁺ **and** Cl⁻

accept in words or dot and cross diagram: must include type and magnitude of charge for each ion

1

electrostatic attraction

allow attraction between opposite charges

1

(b) hydrogen

allow H₂

1

sodium hydroxide

allow NaOH

1

(c) any **one** from, eg:

- people should have the right to choose
- insufficient evidence of effect on individuals
- individuals may need different amounts.

allow too much could be harmful

ignore religious reasons

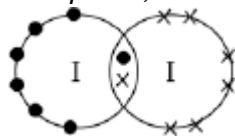
ignore cost

ignore reference to allergies

1

- (d) (i) one bonding pair of electrons

accept dot, cross or e or – or any combination, eg



1

6 unbonded electrons on each atom

1

- (ii) simple molecules

max 2 if incorrect structure or bonding or particles

accept small molecules

accept simple / small molecular structure

1

with intermolecular forces

accept forces between molecules

must be no contradictory particles

1

which are weak **or** which require little energy to overcome – must be linked to second marking point

reference to weak covalent bonds negates second and third marking points

1

- (iii) iodine has no delocalised / free / mobile electrons or ions

1

so cannot carry charge

if no mark awarded iodine molecules have no charge gains 1 mark

1

[14]

5.

answers apply to:

accept diagrams and/or descriptions

carbon dioxide CO₂

ammonia NH₃

methane CH₄

water H₂O

*outer electronic structure of one atom correct **or** needs
correct number of electrons to complete outer shell

1

*outer electronic structure of other atom correct **or** needs
correct number of electrons to complete outer shell

1

*one shared **pair** of electrons (as one covalent bond)

use of ions or reference to ionic bonding negates this mark

1

*outer electronic structure of compound correct **or** each atom now
has a full outer shell/noble gas electron structure

1

[4]

Examiner reports

2.

- (a) This question differentiated well, with the majority of students gaining two or more marks. The best responses were generally succinct and clearly expressed. The ratio of one Mg to two Cl (third marking point) was least frequently awarded. Although the direction of electron transfer was often correct, the number of electrons involved was less often correct. Some students used the terms “ions” and “atoms” interchangeably, limiting their response to a maximum of three marks. A minority of students gained only one mark, often for the final marking point, giving “full outer shell” or “ionic bond”.

Some students suggested that an electron became delocalised because magnesium needed to lose two electrons, but chlorine could only gain one.

Sharing of electrons was sometimes linked with a correct description of electron transfer. These students seemed to think that “sharing” meant to transfer, like sharing out sweets.

- (b) (i) A minority of students knew that the movement of ions is necessary, and that this is why the magnesium chloride must be molten or in aqueous solution for electrolysis to take place. Fewer students indicated that the movement is directional (ie to the electrodes) or that the ions carry charge to gain the second marking point. Some students failed to specify a type of particle which should move, or gave an incorrect type of particle, suggesting that electrons, atoms elements or electrodes would move.
- (ii) Some students clearly understood the processes involved, giving a succinct explanation, gaining all three marks. Most students accessed the first marking point, realising that magnesium is attracted to the negative electrode. Many of these students realised that two electrons were transferred, also gaining the third marking point. The second marking point was least often awarded as there was some confusion between the loss and gain of electrons.
- (iii) Only a minority of students knew the gas produced to be hydrogen. Frequently seen incorrect responses included chlorine, oxygen, carbon dioxide and magnesium oxide or magnesium hydroxide.
- (iv) Incorrect responses often focussed on whether the gas dissolved or whether heating was involved. Only a small number of students gained the mark.
- (v) Only a small number of students gained credit. The most commonly seen errors included adding another element or compound into the equation, eg Mg, CO₂ or H₂O, but a small number of students made a valiant attempt and only omitted the charge on the electron, or made a small error in balancing.
- (c) Many students realised that metals are formed of layers of atoms, but were then unable to describe the movement accurately. Some students listed the properties of metals or suggested that there are weak (intermolecular) forces or bonds between the layers.

3.

- (a) The majority of students were able to state that acid rain is formed and then give a correct consequence of acid rain. Some students simply stated that sulfur dioxide made rain water acidic – it is acidic naturally due to carbon dioxide dissolving in it, or gave vague statements such as it caused “pollution”. A few students blamed sulfur dioxide for causing just about every environmental problem of which they could think.
- (b) (i) A small minority of students got the definitions of oxidation and reduction mixed up, and so despite the equation they were given they tried to say that the copper gained electrons. A few tried to explain oxidation in terms of oxygen gain, using some very inventive chemistry to tie it into the equation provided.
- (ii) This was very often correct. The most common errors were to have the electrons on the wrong side or to have the electrons shown as having a positive charge.
- (iii) This question was demanding. Common errors included suggested non-conductors as the electrolyte (such as distilled water), substances that did not contain copper ions (cryolite and brine were popular) or even molten copper.
- (c) Most students gained the mark for stating that metals contained delocalised electrons, but addressed only the second part of the question, did not describe the bonding, and so did not refer to the positive ions nor the attraction between the positive ions and the electrons. There is some misunderstanding as to the role of electrons in the conduction of electricity, a significant number of students wrote about electrons being able to pass charge on from one electron to the next, not realising that the charge that is moving is the electrons themselves.
- It was very common for students to fill all of the space provided, and so detailed explanations of malleability and melting point were very common. These had nothing to do with the question asked and so gained no marks.
- (d) Only a small minority of students did not realise that the process had something to do with plants. However, as was the case the last time a question covering this part of the specification was asked, it was a common error to state that the ashes left contained copper rather than copper compounds.
- (e) A small minority of students had little idea how to tackle this question, but for many it was a good source of marks. The most common errors were excessive rounding of numbers and careless slips; some of the slips were so obvious that if the candidate had checked their work through they should have been able to spot them. Students would be well advised to put words in their working, or to set it out in a table with headings that make it clear what they are doing because it can be difficult to award any marks for incorrect answers where the working is simply numbers.

4.

- (a) Only a very small minority of students were able to fully describe the structure and bonding without introducing an incorrect term, the most common of which was intermolecular. A large number of students included a description of the losing and gaining of an electron, or included the properties of an ionic compound. Students should be sure to read questions carefully, and answer the question which has been set. Many students included covalent bonding in their response. Some students attempted to include iodine in their descriptions.
- (b) Most students were able to name one of the products of electrolysis of sodium chloride solution, usually hydrogen. Non-creditworthy products frequently seen included sodium, water and carbon dioxide. Sodium oxide was also commonly seen.
- (c) Only a minority of students gained credit. Harmful was commonly given, but not accepted unless qualified to indicate that too much iodine can be harmful. A significant minority of students suggested that iodine should not be added because it is unnatural, even suggesting that it is unnatural because it is a chemical.
- (d) (i) There were many correct responses, with about two thirds of students gaining both marks. Of those gaining 1 mark, most gave the bonding electron pair correctly, but gave an incorrect number of non-bonding electrons. Some gave 2 bonding pairs.
- (ii) Despite some excellent responses, this question proved challenging to many students, many of whom confused the strong covalent bonds within an iodine molecule with the weak intermolecular forces between the molecules. Some students referred to the weak intermolecular forces, but omitted to mention that iodine has simple molecules. There were many references to the reactivity of iodine, and its electron structure, especially the strength of attraction of electrons to the nucleus. Reference to weak covalent bonds negates any mention of intermolecular forces.
- (iii) Most students gained the first marking point by identifying that iodine has no ions or delocalised electrons. However, only a few gained the second marking point for the idea that the ions or delocalised electrons are needed to carry charge. Some students referred to the iodine being liquid, and stated that the particles were too far apart to conduct, and particles have to be closer (as in a solid) to collide and conduct electricity. This was not creditworthy.

5.

A minority of candidates missed the point of this question and failed to describe detailed bonding of any of the compounds mentioned. It was rare to see a covalent bond described as a shared pair of electrons. Diagrammatic responses tended to score better than written descriptive answers. Only a few candidates gave a full account of rock recycling rather *than* the standard rock cycle with igneous, sedimentary and metamorphic rocks.